



Graduation Project — Task Documentation

Person 1 | Authentication + Company Setup

Backend • Frontend

Powered by LLM + RAG + Agentic AI

Overview

You are responsible for everything related to user identity and company onboarding. Every other feature depends on your work — without authentication and KPI setup, nothing else can run. Your deliverable is the security foundation of InsightX AI.

Backend Tasks

ASP.NET Core | JWT | Entity Framework

1. Database Tables to Create

| Table         | Key Columns                                                  |
|---------------|--------------------------------------------------------------|
| Companies     | Id, Name, CreatedAt                                          |
| Users         | Id, Name, Email, PasswordHash, Role, CompanyId, DepartmentId |
| Departments   | Id, Name, CompanyId                                          |
| KPIs          | Id, Name, Threshold, Unit, CompanyId                         |
| RefreshTokens | Id, Token, UserId, ExpiresAt                                 |

2. API Endpoints to Build

| Task                | Details                                   |
|---------------------|-------------------------------------------|
| POST /auth/register | Register a new company + owner account    |
| POST /auth/login    | Return JWT access token + refresh token   |
| POST /auth/refresh  | Issue new access token from refresh token |
| POST /auth/logout   | Revoke refresh token                      |

| Task                 | Details                                        |
|----------------------|------------------------------------------------|
| GET /companies/me    | Return current company profile                 |
| PUT /companies/setup | Save KPIs and Thresholds during onboarding     |
| POST /departments    | Create a new department                        |
| GET /departments     | List all departments for current company       |
| POST /users/invite   | Create a Manager account under current company |
| GET /users           | List all users for current company             |

### 3. Security Rules

- Passwords must be hashed using BCrypt — never store plain text.
- JWT tokens must carry: UserId, CompanyId, DepartmentId, Role.
- Every API must verify CompanyId from the token — users can NEVER access another company's data.
- Role-based authorization: Owner sees everything; Manager sees own department only.
- Refresh tokens expire after 7 days; access tokens after 15 minutes.

## Frontend Tasks

Angular | Bootstrap | Reactive Forms

### Pages to Build

| Task            | Details                                                                         |
|-----------------|---------------------------------------------------------------------------------|
| Login Page      | Email + password form. On success save JWT to localStorage and redirect.        |
| Register Page   | Company name, owner name, email, password. Calls POST /auth/register.           |
| Onboarding Page | After register: add departments, define KPIs and thresholds. Multi-step wizard. |
| Users Page      | Owner only. List users, invite new manager, assign to department.               |
| Profile Page    | Show current user info. Allow password change.                                  |

### Auth Guard

Implement an Angular Route Guard that checks JWT validity before allowing navigation. If token is expired call refresh endpoint silently. If refresh fails redirect to login.

## Example Data in Database

| Id | Name        | Role    | CompanyId | DepartmentId |
|----|-------------|---------|-----------|--------------|
| 1  | Ahmed Samir | Owner   | 1         | NULL         |
| 2  | Mohamed Ali | Manager | 1         | 1            |
| 3  | Sara Hassan | Manager | 1         | 2            |

